

★ Introduction to Azure App Services

Azure App Service is a fully managed Platform-as-a-Service (PaaS) for hosting **web applications, APIs, and background jobs** without worrying about servers, OS patches, load balancers, or deployments.

You focus on **code** → Azure handles **scaling, OS patches, reliability, deployment slots, CI/CD**.

★ Types of Azure App Services

Azure App Service includes **four main offerings**:

1 Web Apps

Used to host:

- ASP.NET / ASP.NET Core
- Node.js
- Python
- Java
- PHP

✓ Designed for **web applications and REST APIs**.

2 API Apps

Same as Web Apps but with API-specific features:

- Swagger/OpenAPI integration
- API versioning
- CORS built-in
- Easy integration with Logic Apps

Use Case: Host backend APIs used by mobile apps, web apps, or microservices.

3 Mobile Apps

Used to build mobile backends:

- Push notifications
- Offline sync

- Mobile authentication (Google, Facebook, AAD)

Use Case: Apps like Swiggy, Ola, Zomato that need reliable mobile push notifications.

Function Apps (Serverless)

Used to run small functions:

- Event-driven (timer triggers, queue triggers)
- Granular billing
- Auto scale to zero

Use Case:

Send emails, process queue messages, background jobs.

★ **How App Service Fits in Azure Architecture**

Component	Purpose
AppService Plan	Provides compute (CPU, RAM, scaling).
AppService	Your web/mobile/API app hosting environment.
Deployment Slots	Deploy safely before swapping to production.
Kudu	Advanced deployment engine for App Services.

★ **Setting Up and Configuring Azure App Services (Step-by-Step Demo)**

Below is a **practical Azure Portal demo** you can follow.

DEMO 1: Create an Azure Web App (ASP.NET Core)

Step 1: Open Azure Portal

 <https://portal.azure.com>

Step 2: Create Web App

- Search → **App Services**
- Click **Create**
- Choose:

- Subscription
- Resource Group (ex: RG-AppServiceDemo)
- Name: mywebapp-demo-2025
- Runtime stack: **.NET 8**
- Region: **Central India**
- App Service Plan:
 - SKU: B1 (Basic)

Click **Review + Create** → **Create**

DEMO 2: Deploy Your ASP.NET Web App

Option 1: Publish from Visual Studio

- Right-click the project → Publish
- Select **Azure → App Service**
- Login → Choose the App → Publish
- ✓ App deployed.

Option 2: Deploy using CI/CD (GitHub Actions)

- Go to App Service → Deployment Center
 - Connect to GitHub repo
 - Azure creates a GitHub workflow file.
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DEMO 3: Configure App Settings

Go to:

→ App Service → Settings → Configuration

Add:

Key: ConnectionStrings:DefaultConnection

Value: <SQL Server connection>

Click **Save** → Auto restart.

DEMO 4: Enable Logging

Go to:

→ Monitoring → App Service Logs

Enable:

- Application Logging: On
 - Log Stream: Live logs
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★ Benefits of Azure App Services (Why companies use them)

Below are **enterprise-level advantages**:

1 Fully Managed Platform

Azure handles:

- OS updates
- Security patches
- Load balancing
- Auto scaling

 You only manage the application code.

2 Automatic Scaling

- Scale Up (increase CPU/RAM)
 - Scale Out (increase instances)
 - Auto scale rules (CPU > 70%)
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Built-in Security

- HTTPS only
- Easy integration with:
 - Azure AD
 - Managed Identities

- SSL Certificates

5 DevOps Support

Supports:

- GitHub Actions
- Azure DevOps
- BitBucket
- Container deployment (Docker)

6 Global Availability

Hosted in **60+ regions**.

★ Real Use Cases of Azure App Services

Use Case	Example
Hosting Websites	Company websites, admin portals
API Hosting	Mobile app backend, microservices
E-commerce	Order APIs, payment processing
Healthcare apps	Patient portals, appointment systems
IoT Dashboards	Data visualization APIs
Education platforms	Learning management systems

★ Azure App Services vs Traditional VMs

Feature	Azure AppService	Traditional VM
Server maintenance	✗ Not needed	✓ Needed
OS updates	Auto	Manual
Deployment	1-click CI/CD	Manual / scripts

Feature	Azure AppService	Traditional VM
Scaling	Auto	Requires additional VMs
Cost	Pay per plan	Pay per VM usage
Security	Managed by Azure	Managed by you
Suitable for	Web apps, APIs	Custom software, databases

★ When to Use What?

Use Azure App Service when:

- ✓ You host a web app or API
- ✓ You want quick deployment
- ✓ You want PaaS (low maintenance)

★ Summary

You learned:

- ✓ What Azure App Services are
- ✓ Types: Web, API, Mobile, Function Apps
- ✓ How to create and deploy an App Service
- ✓ Configuration settings and deployment slots
- ✓ Benefits like scaling, security, CI/CD
- ✓ Use cases
- ✓ App Service vs Traditional VMs